EMBEDDED INTERVIEW FLASHCARDS

Embedded software engineer interview preparation

30 cards — 3 categories

TECHNICAL SKILLS

TECH

RTOS vs Bare Metal

RTOS (FreeRTOS, VxWorks): scheduling, multitasking, priorities.
Bare metal: infinite loop, ISR, full control.
Choice depends on complexity, real-time needs, certification.

TECH

ARM Cortex M0 vs M4

M0: low cost, low power, 2-stage pipeline.
M4: DSP, FPU, 3-stage pipeline, higher performance.
Both use Thumb-2 instruction set.

TECH

CAN Bus — Principle

Differential bus, priority-based arbitration (non-destructive CSMA/CD), standard 11-bit / extended 29-bit frames, max 1 Mbps. Used in automotive, aerospace, industrial.

TECH

BLE vs Bluetooth Classic

BLE: low power, short packets, fast connection.
Classic: higher throughput, audio streaming.
BLE dominates IoT/medical/wearables.

TECH

I2C vs SPI

I2C: 2 wires (SDA/SCL), addressing, multi-master.
SPI: 4 wires (MOSI/MISO/SCK/CS), faster, full-duplex.
SPI for speed, I2C for simplicity.

TECH

MISRA C — Purpose

C coding rules for safety-critical software (automotive, aerospace, medical). Prevents undefined behavior, improves reliability. ~143 mandatory + advisory rules.

TECH

DO-178C — Levels

Aerospace standard for embedded software. 5 levels (A-E). **Level A** = catastrophic (most stringent). Full traceability: requirements → code → tests.

TECH

IEC 62304 — Classes

Medical device software standard. **Class A**: no injury. **Class B**: non-serious injury. **Class C**: death/serious injury. Higher class = more documentation required.

TECH

ISO 15118 — V2G

Vehicle-to-grid charging communication standard. Plug & Charge, PLC (HomePlug Green PHY), TLS authentication, bidirectional energy management.

TECH

Watchdog Timer

Hardware timer that resets the MCU if software fails to refresh it periodically. Protects against hangs and infinite loops. Essential in safety-critical embedded systems.

INTERVIEW QUESTIONS

Q&A

Tell me about your background

20+ years in embedded. Aerospace (Safran, Thales — DO-178), medical (GE Healthcare — IEC 62304), automotive (Valeo), IoT cybersecurity (Enedis/Linky), V2G (E2-CAD — ISO 15118). C/C++, RTOS, embedded Linux.

Q&A

How do you debug a crash on target?

1) Reproduce (logs, watchdog). 2) JTAG/Lauterbach TRACE32 for stack trace. 3) Analyze registers, PC, LR. 4) Memory dump + fault handler. 5) Valgrind if Linux.

Q&A

Experience with embedded testing?

Unit tests (Unity, CUnit), HLT on target, MC/DC coverage (aerospace), integration tests, fuzzing (Enedis), Lauterbach TRACE32 for step-by-step execution.

Q&A

How do you manage memory in embedded?

Static allocation preferred. Memory pools. No malloc in critical code. Separate stack/heap. MPU for protection. Worst-case stack usage analysis.

Q&A

Difference between interrupt and polling?

Interrupt: reactive, low latency, but complex (priorities, reentrancy).
Polling: simple, deterministic, but wastes CPU.
Keep ISRs short → set flag + process in task.

Q&A

Embedded cybersecurity experience?

Enedis: pentest on Linky concentrators, code audit, protocol sniffing (Wireshark, SDR), fuzzing. Security reviews with manufacturers. Kali Linux, Nmap, Scapy.

Q&A

What do you bring to the team?

Cross-sector versatility, fast autonomy, certification rigor, reverse engineering ability. Founder of Workshop-DIY: knowledge sharing, maker mindset.

Q&A

How do you work in a team?

Systematic code reviews, Doxygen documentation, Jira/Polarion for traceability. Clear communication with hardware and system teams. Occasional pair programming.

Q&A

Why are you leaving your current position?

E2-CAD contract completed (March 2026). Looking for a new technical challenge in a demanding environment. Interest in [target company's sector].

Q&A

Your salary expectations?

France: 55-70K EUR/year.
Switzerland: 110-140K CHF/year.
Luxembourg: 75-95K EUR/year.
Open to discussion based on total package.

VOCABULARY EN / FR

VOCAB

Embedded software / Firmware

Logiciel embarque

VOCAB

PCB — Printed Circuit Board

Carte electronique

VOCAB

HAL — Hardware Abstraction Layer

Couche d'abstraction materielle

VOCAB

Device driver

Pilote (driver)

VOCAB

Scheduler

Ordonnanceur

VOCAB

Hard real-time

Temps reel dur

VOCAB

Test bench / Test rig

Banc de test

VOCAB

Commissioning

Mise en service

VOCAB

Requirements specification

Cahier des charges

VOCAB

V-model (development lifecycle)

Cycle en V